Rathi Varshini R

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Computer Science (AI & ML) undergraduate who builds and deploys full-stack, cloud and NLP systems, including a FAISS-based hybrid semantic search engine with RAG. 450+ CodeChef problems solved; LeetCode top 19.74% globally.

EDUCATION

KIT – Kalaignarkarunanidhi Institute of Technology — Coimbatore, India
BE Computer Science Engineering (AI & ML) – CGPA: 8.40/10.0 — Expected graduation: Sep 2028
Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming

EXPERIENCE

Thiranex — Chennai, India
Full Stack Developer Intern — May 2026
- Independently designed, built and deployed four full-stack web apps (portfolio, task manager, e-commerce, blog) with React.js, Node.js and Express REST APIs; owned the pipeline end to end (React on Vercel, Node.js back ends on Render).

DecodeLabs — Lucknow, India
Cloud Computing Intern — May 2026
- Independently provisioned four AWS workloads for assigned business scenarios: S3 static hosting, an EC2 web server, a DynamoDB database and a serverless Lambda API.
- Administered an Ubuntu server over SSH (updates, Apache, services) and validated Lambda behavior in CloudWatch logs.

PROJECTS

Meaning – Semantic Search Engine with Hybrid Retrieval & RAG (AI/ML) | Python, FastAPI, sentence-transformers, FAISS, BM25, scikit-learn, React | Live · GitHub
- Built end to end: a FastAPI back end embedding 26,872 customer-support records with all-MiniLM-L6-v2, searched via a cached FAISS index, plus a React UI comparing semantic and keyword results.
- Implemented hybrid ranking (normalized semantic + BM25 scores), a RAG endpoint that grounds LLM answers in retrieved records, and a t-SNE embedding map.
- Wrote an evaluation harness (Top-1, Top-3, MRR, latency): on a 25-query labeled test set, semantic search reached 100% top-1 accuracy vs 24% for BM25 keyword search (100% vs 5% on paraphrased queries), MRR 1.00 vs 0.27, at ~22 ms per query.

ShopNow – Full Stack E-Commerce | React, Node.js, Express, MongoDB, JWT | Live · GitHub
- Architected a MERN platform with JWT role-based access (Admin/User), a persistent cart, and a catalog with search, category filter, sorting and pagination.
- Implemented a four-stage order workflow (Processing → Delivered) with server-side 18% GST calculation, and built an admin dashboard for products, orders and users.

Deep Learning Projects (AI/ML) | PyTorch, scikit-learn | GitHub
- Implemented three PyTorch models: a 2-16-8-1 ANN (Adam, BCE loss, 200 epochs) for non-linear classification, a 3-block CNN with batch normalization and dropout, and a causal Transformer language model with multi-head self-attention.

AWS Cloud Projects (S3, EC2, DynamoDB, Lambda) | AWS, Node.js, Python (boto3), Linux | Live site · GitHub
- Deployed a React portfolio as a serverless static site on S3 (public-read bucket policy) and launched a t3.micro Ubuntu EC2 instance running Apache, accessed over SSH keys.
- Provisioned a DynamoDB table read via boto3 and a Node.js Lambda behind API Gateway; CloudWatch shows 34.82 ms execution at 128 MB (74 MB used).

FairShare – Group Accountability System (Algorithms & Data) | Python, Flask, SQLite, SQLAlchemy | GitHub
- Designed a weighted fairness-scoring algorithm: task completion 40%, on-time rate 30%, activity 20%, peer ratings 10%.
- Engineered a normalized 7-table schema (SQLAlchemy ORM) with full CRUD, secured with Flask-Login sessions and Werkzeug password hashing.

Ritnezz – Fitness Progressive Web App (Full Stack) | Next.js, TypeScript, Supabase, Tailwind CSS, shadcn/ui | Live · GitHub
- Built a Next.js PWA on Supabase auth and database: weekly workouts with countdown timers and rep tracking, plus a Smart Food Tracker with daily calorie aggregation and analytics.

TECHNICAL SKILLS

Hands-on (used in my projects)
Languages: Python, JavaScript, TypeScript, C++, SQL, Bash
Full Stack: React.js, Next.js, Node.js, Express.js, Flask, FastAPI, REST APIs, JWT, Tailwind CSS
AI/ML: PyTorch, scikit-learn, NLP, sentence-transformers, FAISS, embeddings, RAG
Cloud & Data: AWS (S3, EC2, Lambda, API Gateway, DynamoDB, CloudWatch), MongoDB, SQLite, Supabase, Linux, Git, Postman
Working knowledge (coursework & certifications)
Java, TensorFlow, Docker, PostgreSQL, AWS RDS, Figma, Jira, Agile/Scrum

CERTIFICATIONS

- AWS Academy Graduate – Cloud Foundations (20 hrs) — AWS Academy, 2026
- Oracle Cloud Infrastructure Foundations Associate — SkillSoft, 2026
- Algorithmic Graph Theory and Data Structures (Elite) — NPTEL, 2026
- Applied AI for Customer Review Analysis — Cisco Networking Academy, 2026

COMPETITIVE PROGRAMMING

LeetCode: max rating 1637, top 19.74% globally, 100+ solved | **CodeChef:** max rating 1330 (1 Star), 450+ solved

ACHIEVEMENTS

- **Academic rank:** top 5 CGPA holder (CGPA 8.40/10).
- **Salesforce Agentblazer Champion (2025):** completed 48+ Trailhead badges across platform fundamentals, Agentforce and AI-driven automation.
- **Hackathons:** competed in TN IMPACT, TNWISE 2025 and 2026, and a 24-hour intra-college hackathon.

VOLUNTEERING & LEADERSHIP

- **Student Representative** for 60+ first-year students, the sole liaison between the batch and faculty.
- **Hosted Japanese delegates** at the KIT x H2 Innovation Co., Ltd. (Japan) MoU programme, addressing them in Japanese.
- **Addressed 400+ incoming first-year students** on student companionship in the college auditorium during orientation.